

An Idealized Equinox Hadley Cell

Atmospheric Dynamics

Homework 4, Problem 1

Selected solution, edited for standalone reading

An idealized equinox Hadley cell joins an angular-momentum-conserving tropical atmosphere to radiative-convective equilibrium. In the small-angle approximation, distance from the equator is $y = a\phi$, and the model profiles are

$$\theta_R(y) = \theta_{R0} - \Delta_h \theta (y/a)^2, \quad u(y) = \Omega y^2/a,$$

$$\theta(y) = \theta_0 - \frac{\Omega^2 \Theta_0}{2gHa^2} y^4.$$

Temperature continuity at the edge and the integrated heat balance determine the cell width and the equatorial temperature. The calculation below follows the selected coursework solution; the figure is its original temperature plot.

Radiative-equilibrium wind

Derivation of $u_R(y)$ We first derive $u_R(y)$ based on the thermal wind balance equation:

$$f \frac{\partial u}{\partial z} = -\frac{g}{\Theta_0} \frac{\partial \theta}{\partial y}$$

where f is the Coriolis parameter, g is the acceleration due to gravity, and Θ_0 is the reference potential temperature.

Using the small angle approximation, we have $\sin \phi \approx \phi \approx y/a$:

$$f = 2\Omega \sin \phi \approx 2\Omega \frac{y}{a}.$$

Assuming $u(0) = 0$, the vertical wind shear can be approximated by the wind at u_R divided by the height H :

$$\frac{\partial u}{\partial z} \approx \frac{u_R}{H}.$$

Next, by the radiative-convective equilibrium potential temperature $\theta_R(y)$:

$$\theta_R(y) = \theta_{R0} - \Delta_h \theta \left(\frac{y}{a}\right)^2 \implies \frac{\partial \theta_R}{\partial y} = -2\Delta_h \theta \frac{y}{a^2}.$$

Substituting back into the thermal wind equation gives:

$$\left(2\Omega \frac{y}{a}\right) \left(\frac{u_R}{H}\right) = -\frac{g}{\Theta_0} \left(-2\Delta_h \theta \frac{y}{a^2}\right),$$

$$\frac{\Omega u_R}{aH} = \frac{g\Delta_h \theta}{\Theta_0 a^2}.$$

Rearranging gives

$$u_R(y) = \frac{gH\Delta_h \theta}{a\Theta_0 \Omega}.$$

Numerical evaluation Given the following standard constants:

- $g = 9.81 \text{ m/s}^2$
- $H = 10^4 \text{ m}$
- $\Delta_h \theta = 50 \text{ K}$
- $a = 6.37 \times 10^6 \text{ m}$
- $\Theta_0 = 303 \text{ K}$
- $\Omega = 7.29 \times 10^{-5} \text{ rad/s}$

We plug these values into $u_R(y)$:

$$\begin{aligned} u_R &= \frac{9.81 \times 10^4 \times 50}{(6.37 \times 10^6) \times 303 \times (7.29 \times 10^{-5})} \\ &= \frac{4.9 \times 10^6}{1.4 \times 10^5} \\ &\approx 35 \text{ m/s} \end{aligned}$$

The zonal wind at the tropopause is approximately 34.86 m/s.

Width and equatorial temperature

Solving for y_{HC} Let

$$A := \frac{\Omega^2 \Theta_0}{2gHa^2}, \quad B := \frac{\Delta_h \theta}{a^2}, \quad Y := y_{HC}.$$

Then,

$$\theta(y) = \theta_0 - Ay^4, \quad \theta_R(y) = \theta_{R0} - By^2.$$

Imposing zero net heating in each cell,

$$\begin{aligned} \int_0^{y_{HC}} \theta(y) dy &= \int_0^{y_{HC}} \theta_R(y) dy, \\ \int_0^Y \theta(y) dy &= \int_0^Y \theta_R(y) dy. \end{aligned}$$

That is,

$$\int_0^Y (\theta_0 - Ay^4) dy = \int_0^Y (\theta_{R0} - By^2) dy,$$

hence

$$\begin{aligned} \theta_0 Y - \frac{A}{5} Y^5 &= \theta_{R0} Y - \frac{B}{3} Y^3, \\ \theta_0 - \frac{A}{5} Y^4 &= \theta_{R0} - \frac{B}{3} Y^2. \end{aligned}$$

The continuity at the edge of the Hadley cell gives

$$\theta(Y) = \theta_R(Y),$$

so

$$\theta_0 - AY^4 = \theta_{R0} - BY^2.$$

Subtracting gives

$$AY^4 - BY^2 = \frac{A}{5}Y^4 - \frac{B}{3}Y^2,$$

so

$$\frac{4}{5}AY^4 = \frac{2}{3}BY^2.$$

Therefore,

$$Y^2 = \frac{5}{6} \frac{B}{A}.$$

Substituting for A and B ,

$$y_{HC}^2 = \frac{5}{6} \frac{\Delta_h \theta / a^2}{\Omega^2 \Theta_0 / (2gHa^2)} = \frac{5gH\Delta_h \theta}{3\Omega^2 \Theta_0}.$$

Thus,

$$y_{HC} = \left(\frac{5gH\Delta_h \theta}{3\Omega^2 \Theta_0} \right)^{1/2}.$$

Using all the parameters, we obtain

$$y_{HC} = \left(\frac{5gH\Delta_h \theta}{3\Omega^2 \Theta_0} \right)^{1/2} = \left(\frac{5(9.81)(10^4)(50)}{3(7.29 \times 10^{-5})^2(303)} \right)^{1/2}.$$

Evaluating this gives

$$y_{HC} \approx 2.253 \times 10^6 \text{ m} \approx 2253 \text{ km.}$$

Solving for θ_0 We know

$$\theta_0 = \theta_{R0} - BY^2 + AY^4.$$

Since

$$5B = 6AY^2,$$

we obtain

$$\theta_0 = \theta_{R0} - \frac{1}{6}BY^2.$$

Hence

$$\theta_0 = \theta_{R0} - \frac{1}{6} \frac{\Delta_h \theta}{a^2} \cdot \frac{5gH\Delta_h \theta}{3\Omega^2 \Theta_0},$$

so

$$\theta_0 = \theta_{R0} - \frac{5gH(\Delta_h \theta)^2}{18\Omega^2 \Theta_0 a^2}.$$

Plugging all the parameters, θ_0 becomes

$$\theta_0 = 303 - \frac{5(9.81)(10^4)(50)^2}{18(7.29 \times 10^{-5})^2(303)(6.37 \times 10^6)^2}.$$

Hence

$$\theta_0 \approx 301.96 \text{ K.}$$

Temperature and wind profiles

We now compare the solutions with the radiative–convective equilibrium solution by plotting $\theta(y)$ and $\theta_R(y)$, and then comparing the wind values at the cell edge.

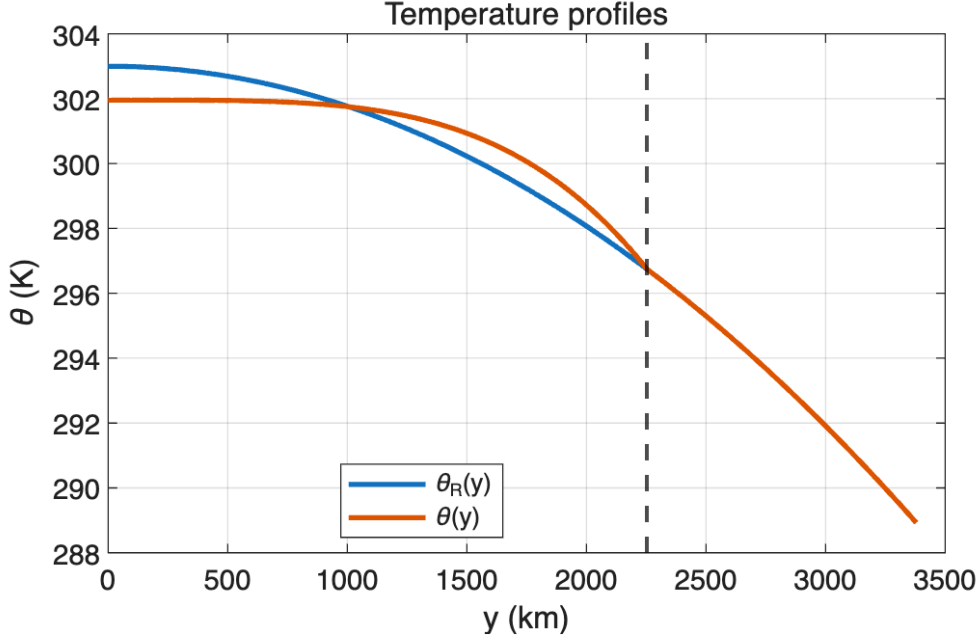


Figure 1: Temperature profiles from the original solution. The dashed line marks the Hadley-cell edge.

The temperature profiles show that $\theta(y)$ matches $\theta_R(y)$ at the edge of the Hadley cell, so the temperature is continuous across the boundary. Inside the cell, the temperature follows the angular-momentum-conserving solution, while outside it follows the radiative equilibrium profile.

The derived wind values at the cell edge are $u(y_{HC}) \approx 58.10$ m/s and $u_R \approx 34.86$ m/s. In contrast to the continuous temperature, the zonal wind has a discontinuity at $y = y_{HC}$. The Hadley-cell wind $u(y)$ increases quadratically in y whereas $u_R(y)$ is constant. As a result, the two do not match at the cell boundary.